

# Migration, Networks, and Religious Choice\*

Timothy Hersey<sup>†</sup>

(Preliminary Draft)

## Abstract

**Abstract:** Over the last half century, affiliation to Pentecostal denominations has grown rapidly, with most of this growth concentrated in developing countries. This paper investigates one economic factor influencing this growth - religious institutions attract members by serving as hosts for economic and social networks. I present new evidence from Ghana, using data from the ISSER Northwestern-Yale Panel Survey (GSPS) and a collected survey of Pentecostal churches in Accra, that religious institutions function as hosts for economic networks, facilitating transfers and referrals. Having established this, I consider the conversion response of a group that particularly value network services - rural-urban migrants. I construct a series of shift-share instruments for migration, using exposure to price shocks in the mining, construction and manufacturing industries. Migration induces conversion both for the migrant, and the remaining household in the village. The process of urbanisation, therefore, may be one important factor explaining the growth of Pentecostalism in urban areas.

---

\*I am indebted to my advisors, Mushfiq Mobarak, Mark Rosenzweig, Jose-Antonio Espin-Sanchez and Ben Polak, for their guidance throughout the many stages of this project. I also thank Lauren Falcao Bergquist, Kevin Donovan, Timothy Guinnane, Costas Meghir, Rohini Pande, Nicholas Ryan, Alvaro Cox, Matthew Schwartzman, Jillian Stallman, and numerous seminar participants at Yale for their insightful comments.

<sup>†</sup>Yale University. Email: [timothy.hersey@yale.edu](mailto:timothy.hersey@yale.edu)

# 1 Introduction

## 1.1 Literature

# 2 Data

## 2.1 Stylised Facts

# 3 Empirical Framework

In this section I seek to establish that migrants, who value network services, convert upon migrating, and convert towards Pentecostal denominations.

For individual  $i$  in community  $d$  at period  $t$ , the effect of migration on affiliation to Religion  $Rel$  is estimated with

$$Rel_{idt} = \beta_R Rel_{i,birth} + \beta_M Mig_{idt} + \beta_X X_{idt} + \alpha_d + \alpha_t + \varepsilon_{idt}$$

where  $Rel_{idt}$  is a dummy variable which equals 1 if individual  $i$  belongs to religion  $Rel$  at time  $t$ ,  $Rel_{i,birth}$  is the religion of individual  $i$  at birth,  $Mig_{idt}$  is a dummy variable which takes the value of 1 if individual  $i$  migrated between periods  $t-1$  and  $t$ ,  $\alpha_d$  is the fixed effect for the community at time 0, and  $\alpha_t$  is the time fixed effect.

This estimation compares migrants with non-migrants from the same community at the baseline period, holding fixed individual characteristics in  $X_{idt}$ .

However, migration may be endogenous to religious choice for several reasons. There may be selection issues in that individuals who are migrants are also more likely to be Pentecostal. For example, migration will be endogenous if younger people are both more likely to demand the services provided by Pentecostal churches and to migrate to the city. Reverse causality would be a problem if conversion assists in the process of migration, if, for example, joining a Pentecostal church connected you to an inter-district network of churches. Finally, there may be unobserved variables that are correlated with both religious affiliation and migration choices. Several papers [Chen, 2010, Ager and Ciccone, 2018, Buser, 2015] have demonstrated

that negative income shocks are likely to increase religiosity. Since income shocks will also affect migration choices, migration will be endogenous if there is no control for these shocks.

### 3.1 Instrument Design

To address these endogeneity concerns, I construct a set of pull instruments using as-good-as-random shocks to industries at potential destination districts. The instrument  $z_{iotk}$  represents the exposure of individual  $i$  from origin district  $o \in \Theta$  at time  $t$  to price shocks in commodity  $k$  in each potential destination district  $d$ . This is a market access style shift-share, where individuals who live closer to districts with high employment potential in industry  $k$  are more exposed to price shocks that affect that industry. The instrument is constructed as

$$z_{iotk} = \phi_{i(t-1)} \sum_{d \in \Theta/o} \frac{\Delta Price_{kt} \times exposure_{kd}}{\tau_{od}}$$

$\Delta Price_{kt}$  is the sum of price innovations between wave  $t - 1$  and wave  $t$ ,  $exposure_{kd}$  is the proportion of workers in industry  $k$  that reside in district  $d$  at baseline (constructed using the 2010 Ghana Census),  $\tau_{od}$  is the distance between the centroids of district  $o$  and  $d$ , and  $\phi_{i(t-1)}$  is a measure of potential individual productivity.

$$\phi_{i(t-1)} = yearsedu_{i,0} \times ageweight_{i,t-1}$$

where  $yearsedu_{i,0}$  is the years of education for individual  $i$  at time 0 and  $ageweight_{i,t-1}$  is a measure of the likelihood of migration based on the age of the migrant at time  $t - 1$ , constructed using the 2010 census.

### 3.2 Instrument Relevance

I exploit random price innovations which affect employment opportunities in one largely rural industry - gold mining, and two urban industries - construction and manufacturing. In 2022, gold made up 47.7% of total exports in Ghana. Work in small scale and artisanal mines is extensive (the number of people working informally in small mines is estimated to be around 1.1 million, McQuilken and Hilson [2016]). Changes in the global price of gold

[width=0.5]Figures/Instrument/instrument<sub>g</sub>old.png

Figure 1: Caption

impacts the profitability of seeking work in gold mining. This in turn induces migration, where households living nearby will be more likely to migrate following this shock. As descriptive evidence for this effect, I regress the share of individuals who report immigrating in a particular year into a gold mining district, according to the 2021 census, on lagged changes in the global gold price series from the FRED database. Results are reported in table ??, and the comovement of series can be seen in figure 3.2.

|                         | <i>Dependent variable:</i> |                       |                     |
|-------------------------|----------------------------|-----------------------|---------------------|
|                         | Prop. of Migrants          | Number of In-Migrants |                     |
|                         | (1)                        | (2)                   | (3)                 |
| Gold Price (USD)/1000   | 0.0001***<br>(0.00004)     | 22.045***<br>(1.699)  |                     |
| Innovation              |                            |                       | 0.018***<br>(0.004) |
| lag(Innovation)         |                            |                       | 0.013***<br>(0.004) |
| District FE             | Yes                        | Yes                   | Yes                 |
| Observations            | 734                        | 734                   | 749                 |
| Adjusted R <sup>2</sup> | 0.467                      | 0.274                 | 0.125               |
| F Statistic             | 38.835***                  | 17.299***             | 6.942***            |

Table 1: Migrants are Adult Males. Regression is at the year-district level.

Secondly I use innovations in the price of construction input (FRED, 2024) multiplied by the US dollar-Ghanaian cedi exchange rate. The construction industry makes up approximately 15% of GDP in Ghana, and employs approximately 420,000 people, especially in urban areas (International Trade Administration, 2023). While approximately 7.9

Thirdly, I use innovations in the price of manufacturing inputs relative to export prices, following the analysis of Kose and Riezman [2001] in examining the effect of trade shocks in Africa. Manufacturing amounts to approximately 11

As in Madhock et al. 2024, I interact the instrument with a measure of individual productivity. The use of this second share exploits individual variation in exposure to the industry shocks. Individuals who are in their early 20s, and who have received more education are more mobile, and therefore are more exposed to the pull-migration shocks.

The two stage least squares framework is the following,

$$Rel_{iodt} = \beta_R Rel_{i,birth} + \beta_M Mig_{iodt} + \sum_k exp_{otk} + \beta_X X_{idt} + \alpha_o + \alpha_t + \varepsilon_{idt}$$

$$Mig_{iodt} = \beta_R Rel_{i,birth} + \sum_k \mu_{1,k} z_{iotk} + \sum_k exp_{otk} + \mu_x X_{iot} + \gamma_o + \gamma_t + v_{iot}$$

where  $exp_{otk} = \Delta Price_{kt} \times exposure_{ko}$  is the direct exposure of district  $o$  to the commodity  $k$  shock.

First stage regressions are reported in tables ———.

### 3.3 Instrument Exogeneity

In order for the instruments to be exogenous, it must be that the instrument only affects religious choice through migration, holding fixed controls. Recent literature in shift-shares has established that consistency can be achieved either through plausible exogeneity of the shifts [Borusyak et al., 2022] or of the shares [Goldsmith-Pinkham et al., 2020]. In this section, I will argue that 1) exposure and distance shares are plausibly exogenous, i.e. that trends over time for districts with more or less exposure are the same absent of migration, and 2) that conditional on controls, shocks to employment opportunities in destination districts are randomly assigned across individuals.

I first consider the exposure of districts to shocks in commodity prices. If estimation were to occur at the district level, we would require that the equivalent of a parallel trends assumption; that trends in Pentecostalism over time across districts are the same absent of the price shock. Without controls, this is unlikely to hold. Districts closer to high density Pentecostal districts are likely to experience differential exposure over time, and districts that are more remote are likely to experience less exposure to religious pluralism. To address these issues I firstly control for region-time fixed effects, so that identification occurs by exploiting differences in the within-district exposure to industry shocks. I also control for population density, and access to Pentecostalism, by including a weighted distance measure to districts with Pentecostalism, similar to the construction of the instrument. I also control

for initial percentage Pentecostal at baseline. To address concerns that exposure to price shocks is correlated across space I directly control for local district exposure to the price shock ( $\Delta Price_{kt} \times exposure_{kd}$ ). There may also be concerns that price shocks in close by districts have spillover effects, either through trade or existing transfer networks. To address this I control for average district income. In figure — below I show that districts with similar weights have similar growth patterns given controls.

Having established that the price-exposure measure provides plausibly random variation across districts, I transform this to an individual exposure, by multiplying by an individual productivity weight - the product of years of education at baseline and an age weight, based on the national likelihood of migrating at age  $a$  based on statistics from the 2000 Ghana census. I control directly for age and years of education level, so that identification occurs through the differential exposure of age-education individuals to the price shock. However many unobservables may be correlated with age and education that predict Pentecostal affiliation. Income, or fertility, for example, will be correlated with both age and income, and may affect religious choice. Using the approach of [Borusyak et al., 2022], the exclusion restriction then rests on the exposure shocks being as-good-as-randomly-assigned given controls and fixed effects, i.e. that exposure to district shocks does not systematically differ by individual productivity weighting  $\phi_{it-1}$ . In figure — below I compare the distribution of shocks for individuals with high and low ageweights, and high and low education levels, after conditioning on controls and fixed effects. I directly control for income, household size, number of births and number of deaths in the household. The distributions appear very similar, so that individuals with different productivity measures are exposed to similar levels of the shock.

More formally, following Borusyak et al. [2022] I run a set of balance tests, regressing potential confounding factors, including education and age weight, against differences in the shock in a pooled cross-section, controlling for region fixed effects. If the shock is uncorrelated with these confounding factors, then we can conclude the shocks are as-good-as-randomly assigned, and that the shock is not arbitrarily picking up comovements with other individual characteristics.

## 4 Migration Results

Since the industry shocks I am using, especially in mining and construction will predominantly affect men, and because the majority of migration for women is for marriage or family purposes, I restrict the sample to male migrants. The estimation here is limited by restrictions in the data - though enumerators made attempts to track down migrants across waves, there is significant attrition. I adjust for this attrition using inverse probability weighting based on observables <sup>1</sup>. The results of the instrument regression are presented in table ??.

|                         | <i>Dependent variable:</i> |                   |                    |                      |                  |                    |
|-------------------------|----------------------------|-------------------|--------------------|----------------------|------------------|--------------------|
|                         | <i>Origin Rural</i>        |                   |                    | <i>Origin Urban</i>  |                  |                    |
|                         | <i>Pentecostal</i>         | <i>Catholic</i>   | <i>Protestant</i>  | <i>Pentecostal</i>   | <i>Catholic</i>  | <i>Protestant</i>  |
|                         | (1)                        | (2)               | (3)                | (4)                  | (5)              | (6)                |
| <i>Migrate</i>          | 0.292*<br>(0.155)          | -0.082<br>(0.120) | -0.246*<br>(0.134) | -0.499***<br>(0.181) | 0.066<br>(0.102) | 0.391**<br>(0.170) |
| Mean                    | 0.35                       | 0.13              | 0.19               | 0.43                 | 0.1              | 0.24               |
| <i>F<sup>Eff</sup></i>  | 16.4316                    | 16.4316           | 16.4316            | 8.2837               | 8.2837           | 8.2837             |
| Region-Wave FE          | Yes                        | Yes               | Yes                | Yes                  | Yes              | Yes                |
| Observations            | 9,462                      | 9,462             | 9,462              | 5,160                | 5,160            | 5,160              |
| Adjusted R <sup>2</sup> | 0.316                      | 0.310             | 0.286              | 0.336                | 0.257            | 0.213              |

Table 2: Contains controls for age, income, education, religion at birth, gender. Includes fixed effects for community at time 0, and Region-wave fixed effects. Errors clustered at the EA level. \*p<0.1; \*\*p<0.05; \*\*\*p<0.01

## 5 Mechansim

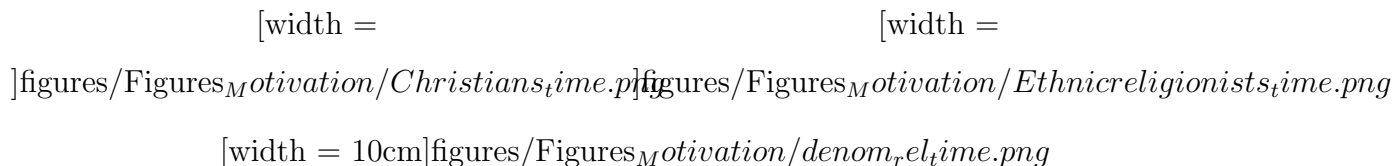
## 6 Discussion

## 7 Conclusion

### Abstract

<sup>1</sup>In most cases, it is known which individuals have migrated, since the household at origin reports their migration event. We do not, however, have information on the current religious affiliation of migrants that are not tracked down

## 8 Motivation



## 9 Context and Setting

Ghana has experienced three key phases of religious transition [Gifford, 2004]. Across the 19th century, traditional Christian denominations (Catholic, Presbyterian, Methodist) set up missions across the south of the country. Some conversion occurred drawn by the services provided by the missions, and the opportunity to align with the colonial elite. The expansion of these missions can be seen in figure 9.



Figure 2: Missions Over Time. Data from Jedwab et al. [2021]

Pentecostalism was introduced in the early 20th century, following Pentecostal revivals in Wales and the western United States. The organisations introduced by Pentecostal missionaries expanded steadily across the 20th century, and today, along with the Catholic church, are the largest religious organisations in the country. The Church of Pentecost, an offshoot of the 1904 Welsh revival, reports a membership of over 3.7 million, or 10.7% of the population of Ghana.

In the 1980s, a new strand of churches emerged - “Charismatic” churches. These churches were typically offshoots of the mainline Pentecostal Churches, with the distinction that the founders were Ghanaians. Charismatic churches proliferate the market today, especially in urban centres. They are many and diverse; the majority are small (under 100 members), though some grow into large, highly marketed organisations of up to 20000 people.

While Charismatic churches are diverse in their doctrine and demographics, [Gifford, 2004] characterises them by several common elements. Firstly, more than traditional forms of Christianity, Charismatic and Pentecostal churches share a core belief in the intervention

of God in every day life, displayed supernaturally through, for example, divine healing, prophecy, and speaking in tongues [The Pew Forum on Religion & Public Life, 2006].

Secondly, Gifford [2004] highlights the emphasis Pentecostal and Charismatic denominations place on success and prosperity. Messages in Charismatic churches often highlight individual agency in creating personal and corporate transformation, promising physical and spiritual rewards for faithful adherence, including escape from poverty [McClendon and Riedl, 2019]. Economic and social success is encouraged, and those who attain this success are often regarded as favoured by God.

Thirdly, while both Catholic and Mainline Protestant churches have invested significantly in economic development, through investments in education and health, political persuasion, human rights advocacy [Gifford, 2004, Wantchekon et al., 2015, Nunn, 2009], many Charismatic churches forgo these services, investing in marketing, radio or tv programming to grow and encourage their congregations. The mainline Pentecostal churches have mostly followed the traditional denominations in pursuing development goals.

Fourthly, the style of worship and music offered in Pentecostal and Charismatic denominations is distinct. Unlike the orderly worship services found in traditional denominations, Pentecostal worship is participatory, exuberant, and personal [Gifford, 2004, Asamoah-Gyadu, 2015, Miller and Yamamori, 2007].

While the traditional denominations and mainline Pentecostal churches have large networks of local branches, and established organisational hierarchies, Charismatic churches typically have one, or a small number of sites, overseen by the same leadership group. The bureaucracy is generally simpler and more centralised.

While the demographics found in Pentecostal and Charismatic churches can vary between organisations and branches, they hold a significant age advantage over the traditional denominations.

For the rest of this description, I will group the mainline Pentecostal church organisations and Charismatic organisations together, noting the differences in their history and organisational structure.

## 10 Existing Literature

The rise in Pentecostal and Evangelical movements around the world has not gone unnoticed. The research mantle was taken up initially by religious studies experts, sociologists and anthropologists before the economics literature began to take note. I will outline here several of the prevailing theories presented in these fields, and connect them to the recent literature in the economics of religion.

1. A much discussed theory for the increasing prevalence of Pentecostalism in Africa is that of cultural similarities. The argument is often presented as follows, initial Christian missions introduced a western form of Christianity that was shallow, and did not resonate with the spiritual experience nor meet the spiritual needs of Africans [Omenyo, 2014]. It wasn't until the formation of African Initiated Churches (AICs) by African converts, that rapid conversion occurred. These churches introduced a form of Christianity that closely resembled many of the practices, rituals and beliefs found in traditional African religions [Asamoah-Gyadu, 2013, 2015, Gifford, 2004, Omenyo, 2014, Miller and Yamamori, 2007]. In economics language, Pentecostalism innovated on the existing product, adapting it to the local setting.

The innovation made by Pentecostal churches may not just be context specific. Miller and Yamamori [2007] hypothesises that the worship service itself may be the good. He asks: “Is this psychological compensation? Or is it full-bodied ecstasy, something to which all human beings might aspire?”.

2. Relatedly, many argue that Pentecostal churches uniquely address the contemporary issues faced by Africans, particularly in urban settings. In his categorisation of Ghanaian Pentecostal churches, Gifford [2004] finds that “these churches have proliferated primarily because they profess to have the answers to Ghana’s real problems”. They address the problems of “joblessness, underemployment, loneliness, and inadequate health care” [Omenyo, 2014] in their practices and in their doctrine. For example, the promotion of miraculous healing is particularly attractive for those who suffer a health shock ([Chesnut, 2003] finds that the most prevalent cause for conversion is sickness), . In their doctrine, many Pentecostal churches preach a form of “the prosperity

gospel”, which promises to reward faithful members with material success [Asamoah-Gyadu, 2013, Wariboko, 2012], a doctrine particularly attractive for individuals facing the struggle of poverty. Similarly, the promotion of holiness in Pentecostal churches encourages men to concentrate their energy on religion and work, and to turn away from alcohol and other vices, which is particularly attractive for ambitious, upwardly mobile individuals [Chesnut, 2003, Omenyo, 2014].

In a summary of the influential work of sociologist David Martin, Wilkinson [2014] states that “Pentecostalism mobilizes the culturally and socially despised of the world, carrying them from the margins to modernity”, through its ability to adapt to the needs of local contexts. Additionally,

Supporting this theory are several empirical studies which demonstrate that religiosity and religious participation increases in response to negative shocks. If Pentecostal churches specialise in providing resources to cope with these events, then shocks may trigger an increase in Pentecostal belief. Ager and Ciccone [2018] find that counties in 19th century United States that are subject to greater common agricultural risk, are associated with a higher level of religiosity, Sinding Bentzen [2019] finds that regions recently exposed to an earthquake tend towards higher religiosity, Buser [2015] finds that unexpected positive income shocks increase religiosity, and the likelihood of attending an evangelical church. Chen and Hungerman [2014] shows that those more exposed to the Indonesian Financial Crisis increased their levels of religiosity and involvement with religious institutions. Binzel and Carvalho [2017] provide evidence that religion acts as psychological insurance in Egypt, providing a coping mechanism when mobility expectations are not met. In their model, individuals substitute time spent in the labour force with religious adherence, reducing the cost of unmet expectations.

Auriol et al. [2020] provide experimental evidence that Pentecostal churches can provide forms of psychological insurance. They offer Pentecostal church members a life insurance product and study their response in a lab-in-the-field dictator game experiment. They find that those who receive the insurance reduce their giving to religious organisations, and do so consistent with the belief that God rewards those who are

financially faithful.

3. Pentecostal churches are unique in their proclivity for marketing, through tv, radio, billboards, the production of music[Gifford, 2004, Asamoah-Gyadu, 2015, Omenyo, 2014]. Chesnut [2003] claims that Pentecostals are not only the biggest advertisers, but the most skilled - pastors in Pentecostal churches share the same social class (language, culture) as the their communities and congregants [Asamoah-Gyadu, 2013]. The participatory nature of Pentecostal churches also lends an “army of amateurs” to provide testimonials, music and to conduct home visits [Chesnut, 2003].
4. Chesnut [2003] argues that Pentecostal churches hold an advantage over other denominations in their organisational structure. Firstly Pentecostal churches reduce costs by starting in houses, or storefronts and democratise their leadership, deferring much of the work to volunteers. Most branches within the Church of Pentecost, for example, do not employ a local pastor. Churches may also concentrate their leadership around the central pastor, reducing bureaucratic effort.

Pentecostal churches also benefit from “unrivalled differentiation” [Chesnut, 2003]. There are no restrictions on the style/ doctrine/ organisational structure of an independent Pentecostal church, and therefore churches are free to specialise and fill niches in the market [Miller and Yamamori, 2007].

In the economics literature, Corbi and Sanches [2021] find that evangelical churches gain market share as a result of tax breaks for religious institutions which favour smaller institutions.

5. Charismatic churches have greater incentives to attract new members, and to extract rents from those members. Typically they are founded by individuals with very little oversight, meaning a high proportion of the income may go to the lead pastor. In comparison, most traditional denominations have a set wage, and are accountable to a church hierarchy. Charismatic leaders may, therefore, be more likely to draw members through one of the already discussed mechanisms - marketing/ creating lively worship services or catering the doctrine to draw the right (wealthy) individuals [Chesnut,

2003]. Seabright [2024] notes that Adam Smith made a similar incentives argument comparing Methodist ministers with Catholic ministers.

6. Some of the sociology literature argues that Pentecostalism reflects broader trends in religious innovation and bureaucratisation. In their summary article on the sociology literature, Wilkinson [2014] highlights the research of David Martin and Margaret Poloma. David Martin argues that the Pentecostal movement occurs in a process of bureaucratisation. As religious groups grow, they must formalise their organisation. This formalisation negatively affects the motivation of followers, and dilutes the incentives of leaders. This creates room for new innovators to enter, who are not constrained by this bureaucracy. In Africa the new innovators are the Pentecostal churches. Poloma argues for a similar process of institutionalisation, with Pentecostalism simply representing a shift in religious providers.
7. A final explanation relates to the role of religions as “platforms” [Seabright, 2024] or networks. Individuals in a church do not just benefit from the services provided by the institution, but from access to the other members of the church. Members may receive a social good from knowing these members, may find potential spouses, or may directly benefit through transfers, job offers, information provision [Murphy et al., 2019] or other typical economic network services [Iyer, 2018].

Mugambi [2023] and Miller and Yamamori [2007] cite urbanisation as a key trigger in the Pentecostal shift. Moving out of the village represented separation from kinship groups and ethnic roots, and displacement from relationships and social norms. This new urban poor was exposed for the first time to a monetary economy, and vast economic inequalities. In the pluralist society of the city, ethnic identities are no longer as strong or necessary for economic advancement. The Pentecostal church is a place where identity is re-established. The Pentecostal church “brings order, stability, and hope to people who are living precarious lives.” Miller and Yamamori [2007]. It does this partly by providing a new group, facilitating new social connections [Mugambi, 2023, Miller and Yamamori, 2007, Asamoah-Gyadu, 2013]. An individual can be surrounded by other upwardly mobile individuals, who share similar values, and who



2. Pentecostalism has expanded its geographic range into rural areas, and increased its concentration particularly in peri-urban, fast growing districts.
3. Religious affiliation is highly geographic - Pentecostal in the south, Muslim in the North, Catholic or Ethnic religions in the far east and far west.
4. There is some correlation between religious affiliation and ethnicity.

## 11.2 Migration

[width =  
]figures/Figures<sub>Motivation</sub>/migrationdestination<sub>2021</sub>.png  
[width = ]figures/Figures<sub>Motivation</sub>/migrationorigin.png  
[width = [width =  
]figures/Figures<sub>MoreMaps</sub>/MigrantPent.png]figures/Figures<sub>MoreMaps</sub>/OutMigrationPentecostal.png

Table 3:

|                              | <i>Dependent variable:</i> |                         |                         |                               |                              |
|------------------------------|----------------------------|-------------------------|-------------------------|-------------------------------|------------------------------|
|                              | <i>Pent<sub>1</sub></i>    | <i>Pent<sub>2</sub></i> | <i>Pent<sub>3</sub></i> | <i>Convert<sub>pent</sub></i> | <i>Convert<sub>any</sub></i> |
|                              | (1)                        | (2)                     | (3)                     | (4)                           | (5)                          |
| <i>HH<sub>moved</sub></i>    | -2.624**<br>(1.191)        | -2.421**<br>(1.165)     | -2.542**<br>(1.288)     | -2.286***<br>(0.849)          | -0.949<br>(1.245)            |
| Controls                     | Yes                        | Yes                     | Yes                     | Yes                           | Yes                          |
| Observations                 | 16,879                     | 16,879                  | 16,879                  | 16,816                        | 16,816                       |
| R <sup>2</sup>               | 0.581                      | 0.505                   | 0.726                   | -0.507                        | 0.707                        |
| Adjusted R <sup>2</sup>      | 0.402                      | 0.294                   | 0.609                   | -1.153                        | 0.581                        |
| <i>F<sup>Effective</sup></i> | 8.5916                     | 8.5916                  | 8.5916                  | 8.5916                        | 8.5916                       |

Table 4: \*

Note: All regressions include wave and household fixed effects, and district\*wave fixed effects. Errors are clustered at the community level. Regression includes controls for income, education, mortality, fertility and marriage. \*p<0.1; \*\*p<0.05; \*\*\*p<0.01

Table 5:

|                               | <i>Dependent variable:</i> |                          |                          |                                |                               |
|-------------------------------|----------------------------|--------------------------|--------------------------|--------------------------------|-------------------------------|
|                               | <i>Pent</i> <sub>1</sub>   | <i>Pent</i> <sub>2</sub> | <i>Pent</i> <sub>3</sub> | <i>Convert</i> <sub>pent</sub> | <i>Convert</i> <sub>any</sub> |
|                               | (1)                        | (2)                      | (3)                      | (4)                            | (5)                           |
| <i>HH</i> <sub>mig</sub>      | 0.158**<br>(0.076)         | -0.095<br>(0.095)        | 0.243**<br>(0.111)       | 0.135**<br>(0.063)             | -0.092<br>(0.137)             |
| Controls                      | Yes                        | Yes                      | Yes                      | Yes                            | Yes                           |
| Observations                  | 16,879                     | 16,879                   | 16,879                   | 16,816                         | 16,816                        |
| R <sup>2</sup>                | 0.778                      | 0.679                    | 0.808                    | 0.375                          | 0.720                         |
| Adjusted R <sup>2</sup>       | 0.684                      | 0.541                    | 0.726                    | 0.108                          | 0.600                         |
| <i>F</i> <sup>Effective</sup> | 65.9259                    | 65.9259                  | 65.9259                  | 65.9259                        | 65.9259                       |

Table 6: \*

Note: All regressions include wave and household fixed effects, and district\*wave fixed effects. Errors are clustered at the community level. Regression includes controls for income, education, mortality, fertility and marriage. \*p<0.1; \*\*p<0.05; \*\*\*p<0.01

### Notes

1. The proportion of Pentecostals in a district is correlated with its migration rate, and to a lesser extent, its out-migration rate.
2. Household movement is negatively associated with Pentecostalism and conversion.
3. Individual migration leads to increases in the number of Pentecostals in the household, and increases conversions to Pentecostalism.

## 11.3 Transfers

Table 7:

|                     | <i>Dependent variable:</i> |                            |                          |
|---------------------|----------------------------|----------------------------|--------------------------|
|                     | Total In Transfers         | From Non-relatives         | From Relatives           |
|                     | (1)                        | (2)                        | (3)                      |
| Ethnic              | −1,539.390***<br>(456.342) | −173.397<br>(420.181)      | −424.375<br>(309.261)    |
| Muslim              | −1,451.312***<br>(314.608) | −82.866<br>(281.472)       | −484.256**<br>(218.151)  |
| Pentecostal         | −558.642**<br>(262.947)    | 248.559<br>(234.169)       | −258.595<br>(181.412)    |
| Traditional         | −50.773<br>(293.961)       | 52.364<br>(263.039)        | 91.824<br>(202.646)      |
| log(income)         | −68.712***<br>(21.876)     | −14.420<br>(18.945)        | −11.993<br>(15.337)      |
| Num Male Adults     | −48.161<br>(125.272)       | 6.392<br>(109.031)         | 54.827<br>(87.972)       |
| Num Children        | −6.826<br>(58.789)         | −21.653<br>(51.726)        | −100.459**<br>(41.050)   |
| Num Elderly         | 380.898***<br>(136.714)    | −809.984***<br>(126.558)   | 1,154.187***<br>(94.347) |
| Num Female Adults   | 142.137<br>(119.281)       | 116.221<br>(104.595)       | −200.035**<br>(83.303)   |
| log(popn density)   | 233.011**<br>(95.598)      | 139.792*<br>(83.940)       | 134.711**<br>(66.564)    |
| urban percentage    | 2,192.245***<br>(513.516)  | −180.281<br>(449.489)      | −6.546<br>(358.131)      |
| logSigma            | 8.446***<br>(0.013)        | 8.083***<br>(0.025)        | 8.017***<br>(0.016)      |
| Constant            | 1,067.808**<br>(468.072)   | −2,373.743***<br>(421.575) | −368.270<br>(326.019)    |
| Observations        | 3,189                      | 3,189                      | 3,189                    |
| Log Likelihood      | −31,432.130                | −10,079.080                | −20,594.500              |
| Akaike Inf. Crit.   | 62,890.260                 | 20,184.150                 | 41,215.000               |
| Bayesian Inf. Crit. | 62,969.140                 | 20,263.030                 | 41,293.880               |

Table 9:

|                     | <i>Dependent variable:</i> |                            |                            |
|---------------------|----------------------------|----------------------------|----------------------------|
|                     | Out Transfers              | To Non-relatives           | To Relatives               |
|                     | (1)                        | (2)                        | (3)                        |
| Ethnic              | −1,027.937***<br>(347.001) | −868.724***<br>(298.851)   | −744.198**<br>(336.087)    |
| Muslim              | −784.749***<br>(229.141)   | −558.487***<br>(191.101)   | −778.222***<br>(222.049)   |
| Pentecostal         | 88.922<br>(196.014)        | 36.835<br>(159.262)        | 88.875<br>(188.406)        |
| Traditional         | 586.339***<br>(221.951)    | 544.568***<br>(177.204)    | 252.062<br>(214.123)       |
| log(income)         | 174.755***<br>(15.722)     | 35.633***<br>(12.862)      | 156.924***<br>(15.075)     |
| Num Male Adults     | 322.579***<br>(88.274)     | 301.683***<br>(69.366)     | 45.484<br>(86.246)         |
| Num Children        | −164.103***<br>(40.956)    | −28.972<br>(32.207)        | −149.035***<br>(40.038)    |
| Num Elderly         | −1,158.913***<br>(108.677) | 54.513<br>(83.894)         | −1,412.392***<br>(111.133) |
| Num Female Adults   | −341.373***<br>(87.253)    | 138.275**<br>(68.101)      | −179.161**<br>(84.661)     |
| log(popn density)   | −10.821<br>(71.330)        | 33.127<br>(59.345)         | −35.088<br>(68.021)        |
| urban percentage    | 490.972<br>(387.103)       | −281.159<br>(318.875)      | 819.662**<br>(370.558)     |
| logSigma            | 8.319***<br>(0.014)        | 7.870***<br>(0.025)        | 8.209***<br>(0.017)        |
| Constant            | −1,045.761***<br>(342.797) | −3,159.762***<br>(294.379) | −1,681.641***<br>(329.225) |
| Observations        | 5,797                      | 5,797                      | 5,797                      |
| Log Likelihood      | −29,801.370 <sup>17</sup>  | −10,850.390                | −22,177.700                |
| Akaike Inf. Crit.   | 59,628.740                 | 21,726.780                 | 44,381.390                 |
| Bayesian Inf. Crit. | 59,715.380                 | 21,813.420                 | 44,468.040                 |

Notes (These Regressions are not Great - should be redone)

1. Ethnicists, Muslims, and Pentecostals tend to receive less in transfers in comparison to Catholics, Muslims receive less from relatives.
2. Ethnicists and Muslims give less transfers, to both relatives and non-relatives. Traditional Christians give more transfers, especially to non-relatives (this could be an indication that they rely less on the formal institution).

## 11.4 Services Received

|             | N    | Support | $Support_{church}$ | $Support_{Network}$ | Money Support |
|-------------|------|---------|--------------------|---------------------|---------------|
| Catholic    | 756  | 0.07    | 0.06               | 0.03                | 0.03          |
| Muslim      | 1276 | 0.03    | 0.03               | 0.02                | 0.00          |
| None        | 69   | 0.09    | 0.07               | 0.03                | 0.03          |
| Pentecostal | 2215 | 0.10    | 0.08               | 0.06                | 0.05          |
| TradChrist  | 1072 | 0.08    | 0.06               | 0.04                | 0.04          |
| Ethnic      | 356  | 0.03    | 0.02               | 0.02                | 0.01          |

Table 11: Services Received by Members of Religious institutions in Rural Areas

|             | N   | Support | $Support_{church}$ | $Support_{Network}$ | Money Support |
|-------------|-----|---------|--------------------|---------------------|---------------|
| Catholic    | 95  | 0.07    | 0.05               | 0.05                | 0.03          |
| Muslim      | 118 | 0.08    | 0.08               | 0.03                | 0.03          |
| None        | 3   | 0.33    | 0.33               | 0.33                | 0.00          |
| Pentecostal | 665 | 0.13    | 0.12               | 0.07                | 0.06          |
| TradChrist  | 301 | 0.14    | 0.11               | 0.07                | 0.06          |
| Ethnic      | 5   | 0.00    | 0.00               | 0.00                | 0.00          |

Table 12: Services Received by Members of Religious institutions in Urban Areas

Notes

1. Pentecostals are more likely to report receiving services from their church and their church network.
2. Services received from the church are more common in urban areas.
3. Churches report giving extensive support to their members. This support takes a variety of forms; material support, help with jobs and housing, training.

Table 13:

|                         | <i>Dependent variable:</i> |
|-------------------------|----------------------------|
|                         | Received Support           |
| log(Pop Density)        | 0.007**<br>(0.003)         |
| Muslim                  | -0.020*<br>(0.010)         |
| None                    | -0.024*<br>(0.014)         |
| Other                   | -0.053***<br>(0.009)       |
| Pentecostal             | 0.035***<br>(0.009)        |
| TradChrist              | 0.018*<br>(0.011)          |
| Ethnic                  | -0.035***<br>(0.011)       |
| Constant                | -0.018<br>(0.024)          |
| Observations            | 8,836                      |
| R <sup>2</sup>          | 0.017                      |
| Adjusted R <sup>2</sup> | 0.016                      |
| Residual Std. Error     | 0.244 (df = 8824)          |

Table 14: Note: Controls for Age,  $Age^2$ , Gender, log(income). \*p<0.1; \*\*p<0.05; \*\*\*p<0.01

|                                   | Number Churches | Mean    | Median | Std     | Min | Max  |
|-----------------------------------|-----------------|---------|--------|---------|-----|------|
| Number of Transfers               |                 |         |        |         |     |      |
| Monetary                          | 31              | 17.15   | 12     | 17.06   | 1   | 80   |
| Food                              | 26              | 17.54   | 10     | 19.08   | 2   | 80   |
| Clothing                          | 22              | 16.25   | 10     | 20.93   | 1   | 80   |
| Size of Transfers                 |                 |         |        |         |     |      |
| Monetary                          | 31              | 1749.17 | 475    | 2324.41 | 100 | 8000 |
| Food                              | 26              | 765.42  | 100    | 1484.98 | 50  | 5000 |
| Clothing                          | 22              | 577     | 100    | 1144.35 | 50  | 5000 |
| Other Network Services            |                 |         |        |         |     |      |
| Helped find job                   | 28              | 15      | 7.50   | 26.22   | 2   | 135  |
| Helped find housing               | 26              | 8.46    | 5      | 12.82   | 1   | 60   |
| Business Related Services         |                 |         |        |         |     |      |
| Business Training (Participation) | 16              | 45.12   | 30     | 46.24   | 3   | 200  |
| Skills Training (Participation)   | 16              | 39.31   | 28.50  | 40.66   | 4   | 170  |
| Businesses Advertised             | 25              | 8.48    | 5      | 9.31    | 1   | 40   |

Table 15: Exploratory survey of Pentecostal churches in Accra: Churches provide network services

## 11.5 Who is Pentecostal?

Table 16:

|                                 | <i>Dependent variable:</i> |                      |                      |
|---------------------------------|----------------------------|----------------------|----------------------|
|                                 | Religious Activities       | Religious Services   | Pray Freq.           |
|                                 | (1)                        | (2)                  | (3)                  |
| Muslim                          | 0.491***<br>(0.075)        | 1.282***<br>(0.053)  | 0.403***<br>(0.038)  |
| None                            | -1.756***<br>(0.138)       | -2.565***<br>(0.097) | -1.103***<br>(0.069) |
| Other                           | 0.419<br>(0.693)           | 0.209<br>(0.486)     | -0.089<br>(0.348)    |
| Pentecostal                     | 0.374***<br>(0.067)        | 0.147***<br>(0.047)  | 0.195***<br>(0.034)  |
| TradChrist                      | 0.188**<br>(0.075)         | 0.018<br>(0.052)     | 0.257***<br>(0.038)  |
| Ethnic                          | -0.875***<br>(0.103)       | -1.656***<br>(0.072) | -1.847***<br>(0.052) |
| Constant                        | -6.294***<br>(0.059)       | -5.209***<br>(0.041) | -3.536***<br>(0.029) |
| Observations                    | 8,920                      | 8,920                | 8,920                |
| R <sup>2</sup>                  | 0.051                      | 0.258                | 0.233                |
| Adjusted R <sup>2</sup>         | 0.051                      | 0.258                | 0.232                |
| Residual Std. Error (df = 8913) | 1.953                      | 1.369                | 0.982                |
| F Statistic (df = 6; 8913)      | 80.644***                  | 517.560***           | 451.058***           |

Table 17: Note: \*p&lt;0.1; \*\*p&lt;0.05; \*\*\*p&lt;0.01

|             | Tithe | Don't tithe |
|-------------|-------|-------------|
| Catholic    | 0.46  | 0.54        |
| Muslim      | 0.18  | 0.82        |
| None        | 0.11  | 0.89        |
| Other       | 0.33  | 0.67        |
| Pentecostal | 0.63  | 0.37        |
| TradChrist  | 0.66  | 0.34        |
| Ethnic      | 0.03  | 0.97        |

Table 18:

|                     | <i>Dependent variable:</i> |
|---------------------|----------------------------|
|                     | Religious Giving           |
| Ethnic              | −294.264***<br>(53.740)    |
| Muslim              | 35.558<br>(22.963)         |
| Pentecostal         | 43.301**<br>(20.188)       |
| Traditional         | 74.360***<br>(22.528)      |
| log(income)         | 4.713***<br>(1.576)        |
| Num Male Adults     | 3.341<br>(8.954)           |
| Num Children        | −8.234**<br>(4.043)        |
| Num Elderly         | 1.117<br>(10.627)          |
| Num Female Adults   | 17.363**<br>(8.588)        |
| log(popn density)   | 23.722***<br>(3.965)       |
| logSigma            | 5.833***<br>(0.022)        |
| Constant            | −476.273***<br>(32.716)    |
| Observations        | 5,794                      |
| Log Likelihood      | −10,951.850                |
| Akaike Inf. Crit.   | 21,927.710                 |
| Bayesian Inf. Crit. | 22,007.680                 |

Table 19: Note: Censored regression with controls for family structure, and some district characteristics. \*p<0.1; \*\*p<0.05; \*\*\*p<0.01

|                                              | None | Christian | Muslim | Traditional |
|----------------------------------------------|------|-----------|--------|-------------|
| Agreed with beliefs of religion/denomination | 0.42 | 0.17      | 0.22   | 0.57        |
| Distance                                     | 0.05 | 0.08      | 0.09   | 0.00        |
| Life cycle or family reasons                 | 0.42 | 0.48      | 0.49   | 0.35        |
| The community                                | 0.00 | 0.07      | 0.04   | 0.03        |
| The economic/material support                | 0.00 | 0.02      | 0.00   | 0.00        |
| The leaders/ministers                        | 0.00 | 0.05      | 0.03   | 0.01        |
| The worship services                         | 0.11 | 0.11      | 0.09   | 0.04        |

Table 20: Reasons for switching religion out of birth religion. Religion at birth is on the x-axis.

|                                              | None | Christian | Muslim | Traditional |
|----------------------------------------------|------|-----------|--------|-------------|
| Agreed with beliefs of religion/denomination | 0.74 | 0.36      | 0.44   | 0.77        |
| Distance                                     | 0.00 | 0.06      | 0.04   | 0.00        |
| Life cycle or family reasons                 | 0.21 | 0.38      | 0.31   | 0.17        |
| The community                                | 0.00 | 0.07      | 0.06   | 0.03        |
| The economic/material support                | 0.00 | 0.02      | 0.00   | 0.00        |
| The leaders/ministers                        | 0.00 | 0.04      | 0.03   | 0.02        |
| The worship services                         | 0.05 | 0.08      | 0.12   | 0.02        |

Table 21: Reasons for switching to current religion. Religion at birth is on the x-axis.

## Notes

1. Pentecostals tend to attend services and activities more frequently than Catholics (and Traditional Christians). Muslims are the most religious on all measures. Pentecostals pray more than Catholics, but less than traditional Christians.
2. Pentecostals and traditional Christians are the most likely to tithe (give a fixed percentage of income).
3. Pentecostals give more to their church than Catholics, muslims and ethnicists. Traditional Christians, however, give the most (controlling for income and household structure)
4. By far the most cited reason for changing religions or denominations is “life cycle or family reasons”, particularly for those who were born Christian. The second most cited reason is beliefs.

## 12 Conclusions

Much of the evidence presented supports the platform argument. We can see in the services information that people are receiving economic services from their religious institution, in line with belonging to a religious network.

If the transfers information holds, it would seem that Pentecostal households are participating less in transfers than their other Christian counterparts, indicating that either they are more independent, or they may be relying on more formal support.

The association between Pentecostalism and migration, and out-migration also supports this explanation, as religious networks substitute for traditional ethnic ones.

Most people change religions for life-cycle purposes, in accord with different economic needs at different stages of life.

There are a few issues with this explanation exclusively 1) While it seems clear that indeed churches are functioning as networks, the causality is not clear. The evidence thus far does not show that the need/demand for a network induces conversion. It may be that this is a

secondary good that is attained. 2) Within the network story, it is not clear what specifically would make the network attractive. It could be economic services, job prospects, marriage prospects, a social good, the desire to be around people like yourself. It could also be that Pentecostal churches charge higher costs, and so those entering receive greater benefits, consistent with the club goods model. 3) I cannot rule out any of the other stories with the evidence presented here entirely. (1) and (2) together represent improvements in the religion technology, which could be especially targeted at migrants. While the reasons for switching indicate this may not be the primary reason, this is only suggestive. The presence of network goods would indicate that this cannot be the only reason. If a church could extract rent from the members and give nothing in return, they would do so. If the social component were not important, we would not see the level of involvement in volunteering and other church activities that are primarily social (women's groups/ men's groups). There is not support currently for (6), where we would see reduction in affiliation as the size of the organisation grows, though some work could be done to show this more completely. (3), (4) and (5) together help to explain why Pentecostalism may have an advantage over other denominations. Marketing, and persuasion are only effective if there is underlying demand. (3) should be testable with information on technology spread. (5) would be testable with information on organisation structure. Given the success of large organisations with many volunteers in the Pentecostal church, this story cannot be the dominant one. (4) likely helps to explain part of the Pentecostal advantage, both in cost reduction, and in the ability to meet the demand for the religious good (i.e. moving from an oligopoly market to a perfect competition one). The churches with lower costs will have advantage in the perfect competition setting. One way to test this may be to look at property prices by region, and to see how successful Pentecostal churches are comparatively to traditional ones.

## References

- Philipp Ager and Antonio Ciccone. Agricultural risk and the spread of religious communities. *Journal of the European Economic Association*, 16(4):1021–1068, 2018.
- J Kwabena Asamoah-Gyadu. *Contemporary pentecostal Christianity: Interpretations from*

- an African context*. Wipf and Stock Publishers, 2013.
- J Kwabena Asamoah-Gyadu. *Sighs and signs of the Spirit: Ghanaian perspectives on Pentecostalism and renewal in Africa*. Regnum Africa, 2015.
- Emmanuelle Auriol, Julie Lassebie, Amma Panin, Eva Raiber, and Paul Seabright. God insures those who pay? formal insurance and religious offerings in ghana. *The Quarterly Journal of Economics*, 135(4):1799–1848, 2020.
- Christine Binzel and Jean-Paul Carvalho. Education, social mobility and religious movements: The islamic revival in egypt. *The Economic Journal*, 127(607):2553–2580, 2017.
- Kirill Borusyak, Peter Hull, and Xavier Jaravel. Quasi-experimental shift-share research designs. *The Review of economic studies*, 89(1):181–213, 2022.
- Thomas Buser. The effect of income on religiousness. *American Economic Journal: Applied Economics*, 7(3):178–95, 2015.
- Daniel L Chen. Club goods and group identity: Evidence from islamic resurgence during the indonesian financial crisis. *Journal of political Economy*, 118(2):300–354, 2010.
- Daniel L Chen and Daniel M Hungerman. Economics, religion, and culture: A brief introduction. *Journal of Economic Behavior and Organization*, 104(1-3), 2014.
- R Andrew Chesnut. *Competitive spirits: Latin America’s new religious economy*. Oxford University Press, 2003.
- Raphael Corbi and Fabio A Miessi Sanches. The political economy of pentecostalism: A dynamic structural analysis. Technical report, 2021.
- André Droogers. The cultural dimension of pentecostalism. *The Cambridge Companion to Pentecostalism*, pages 195–214, 2014.
- Matthew Engelke. Past pentecostalism: notes on rupture, realignment, and everyday life in pentecostal and african independent churches. *Africa*, 80(2):177–199, 2010.

- Paul Gifford. *Ghana's new Christianity: Pentecostalism in a globalizing African economy*. Indiana University Press, 2004.
- Paul Goldsmith-Pinkham, Isaac Sorkin, and Henry Swift. Bartik instruments: What, when, why, and how. *American Economic Review*, 110(8):2586–2624, 2020.
- Laurence R Iannaccone. Sacrifice and stigma: Reducing free-riding in cults, communes, and other collectives. *Journal of political economy*, 100(2):271–291, 1992.
- Sriya Iyer. *The economics of religion in India*. Harvard University Press, 2018.
- Remi Jedwab, Felix Meier zu Selhausen, and Alexander Moradi. Christianization without economic development: Evidence from missions in ghana. *Journal of Economic Behavior & Organization*, 190:573–596, 2021.
- M Ayhan Kose and Raymond Riezman. Trade shocks and macroeconomic fluctuations in africa. *Journal of development Economics*, 65(1):55–80, 2001.
- Gwyneth H McClendon and Rachel Beatty Riedl. *From PEWS to politics: Religious sermons and political participation in Africa*. Cambridge University Press, 2019.
- James McQuilken and Gavin Hilson. Artisanal and small-scale gold mining in ghana. *Evidence to inform an 'action dialogue'*. IIED, London, 2016.
- Donald E Miller and Tetsunao Yamamori. *Global Pentecostalism: The new face of Christian social engagement*. Univ of California Press, 2007.
- Kyama Mugambi. Student movements and spiritual identity in the growth of pentecostalism in kenya. In *The Pentecostal World*, pages 82–94. Routledge, 2023.
- David MA Murphy, Vesall Nourani, and David R Lee. Chatting at church: Information diffusion through religious networks. *Review of Economics and Statistics*, pages 1–46, 2019.
- Nathan Nunn. Christians in colonial africa. *Unpublished manuscript*, 2009.

- Cephas N Omenyo. African pentecostalism. *The Cambridge Companion to Pentecostalism*, pages 132–151, 2014.
- Paul Seabright. *The Divine Economy: How Religions Compete for Wealth, Power, and People*. Princeton University Press, 2024.
- Jeanet Sinding Bentzen. Acts of god? religiosity and natural disasters across subnational world districts. *The Economic Journal*, 129(622):2295–2321, 2019.
- The Pew Forum on Religion & Public Life. Spirit and power: A 10-country survey of pentacostals, 2006.
- Leonard Wantchekon, Marko Klašnja, and Natalija Novta. Education and human capital externalities: evidence from colonial benin. *The Quarterly Journal of Economics*, 130(2): 703–757, 2015.
- Nimi Wariboko. Pentecostal paradigms of national economic prosperity in africa. In *Pentecostalism and prosperity: The socio-economics of the global charismatic movement*, pages 35–59. Springer, 2012.
- Michael Wilkinson. 11 sociological narratives and the sociology of pentecostalism. *The Cambridge Companion to Pentecostalism*, page 215, 2014.

## A Instrument Estimation

### A.1 Baseline Equation

Table 22:

|                                                      | <i>Dependent variable:</i> |
|------------------------------------------------------|----------------------------|
|                                                      | Pentecostal                |
| Age                                                  | −0.002***<br>(0.0002)      |
| I(Age^2)                                             | −0.00001***<br>(0.00000)   |
| SexFemale                                            | 0.046***<br>(0.001)        |
| Ethnicity                                            | −0.187***<br>(0.009)       |
| EthnicityAkan                                        | −0.021***<br>(0.002)       |
| EthnicityAll Other Tribes (Hausa, Baribari, Zabrama) | −0.335***<br>(0.005)       |
| EthnicityGa-Dangme                                   | 0.032***<br>(0.003)        |
| EthnicityGrusi                                       | −0.225***<br>(0.004)       |
| EthnicityGuan                                        | −0.135***<br>(0.004)       |
| EthnicityGurma                                       | −0.138***<br>(0.003)       |
| EthnicityMande                                       | −0.352***<br>(0.005)       |
| EthnicityMole-Dagbani                                | −0.225***<br>(0.002)       |
| Married_red                                          | 0.063***<br>(0.001)        |
| LiterateYes, read and write                          | −0.001<br>(0.002)          |
| LiterateYes, read only                               | −0.003<br>(0.005)          |
| Primary                                              | 0.020***<br>(0.002)        |
| Secondary                                            | 0.016***<br>(0.002)        |
| Tertiary                                             | −0.012***<br>(0.003)       |
| Cert                                                 | −0.004<br>(0.003)          |
| bprot                                                | −0.140***<br>(0.017)       |
| bislam                                               | 0.067***<br>(0.008)        |
| btrad                                                | 0.073***<br>(0.015)        |
| bpent                                                | 0.716***<br>(0.009)        |
| birth_rural                                          | 0.012***<br>(0.002)        |

## B Results

In the following results, I use three different definitions of “Pentecostal”. *Pentecostal*<sub>1</sub> is the sum of household members who identify themselves as “Pentecostal”, or list an organisation which is traditionally recognised as a Pentecostal institution (The Church of Pentecost, Assemblies of God, The Apostolic Church). *Pentecostal*<sub>2</sub> is the sum of household members who identify themselves as “Charismatic”, or list organisations that are not part of large established institutions (e.g., Action Chapel International, Lighthouse Chapel International). *Pentecostal*<sub>3</sub> pools these groups together. *Convert*<sub>pent</sub> counts the number of household members that have converted to Pentecostal between waves (using the third definition of Pentecostal), i.e. were observed as Pentecostal in wave  $t$ , but were not Pentecostal in wave  $t - 1$ . *Convert*<sub>any</sub> counts the number of household members who change between any pair of religious denominations between waves.

I use observed migration in the Ghana Panel between waves i.e., an individual has migrated if they are no longer part of the household and they are reported to have moved outside the community. I also consider household movement (where the whole household relocates to a different community between waves). I define  $t$  to represent the waves of the Ghana panel (2009/2010, 2013/2014, 2017/2018, 2021/2022).

I report the Montiel Olea & Pflueger (2013) F-effective statistic in each of the following regression tables. This modified F statistic allows for settings with non-homoskedastic errors, which I compare with the rule of thumb value of 10 recommended by Andrew, Stock & Sun (2018).

### B.1 Household Move

- Household movement induces higher levels of Pentecostalism in the household.
  - i.e. in periods where the household moves, they have more Pentecostal members
- Household movement appears to lead to lower levels of conversion across religions, with no effect on denominational conversion.
  - Moving households have less religious converts in the period they move in.

Table 23:

|                                | <i>Dependent variable:</i> |                         |                         |                                |                              |                               |
|--------------------------------|----------------------------|-------------------------|-------------------------|--------------------------------|------------------------------|-------------------------------|
|                                | <i>Pent<sub>1</sub></i>    | <i>Pent<sub>2</sub></i> | <i>Pent<sub>3</sub></i> | <i>Convert<sub>denom</sub></i> | <i>Convert<sub>rel</sub></i> | <i>Convert<sub>Pent</sub></i> |
|                                | (1)                        | (2)                     | (3)                     | (4)                            | (5)                          | (6)                           |
| HH log(income)                 | 0.001<br>(0.002)           | −0.0005<br>(0.003)      | 0.0004<br>(0.004)       | 0.003<br>(0.002)               | 0.003<br>(0.002)             | 0.001<br>(0.001)              |
| <i>Num<sub>Edu&gt;0</sub></i>  | 0.069**<br>(0.030)         | 0.227***<br>(0.038)     | 0.322***<br>(0.049)     | 0.224***<br>(0.041)            | 0.134***<br>(0.038)          | 0.053**<br>(0.021)            |
| <i>Num<sub>Primary</sub></i>   | 0.010<br>(0.023)           | −0.084***<br>(0.025)    | −0.049<br>(0.034)       | 0.024<br>(0.033)               | −0.006<br>(0.030)            | −0.029*<br>(0.017)            |
| <i>Num<sub>Secondary</sub></i> | −0.066**<br>(0.031)        | −0.003<br>(0.031)       | −0.057<br>(0.049)       | −0.090**<br>(0.039)            | −0.044<br>(0.035)            | 0.020<br>(0.016)              |
| <i>Num<sub>Tertiary</sub></i>  | 0.146**<br>(0.066)         | −0.003<br>(0.085)       | 0.134<br>(0.104)        | 0.127<br>(0.093)               | 0.003<br>(0.078)             | −0.033<br>(0.045)             |
| <i>Num<sub>Prof.Deg.</sub></i> | 0.035<br>(0.042)           | −0.063<br>(0.052)       | 0.0005<br>(0.073)       | 0.089*<br>(0.051)              | −0.079*<br>(0.044)           | −0.034<br>(0.025)             |
| <i>Num<sub>Married</sub></i>   | 0.088***<br>(0.019)        | 0.085***<br>(0.021)     | 0.243***<br>(0.031)     | 0.133***<br>(0.022)            | 0.031<br>(0.021)             | 0.004<br>(0.011)              |
| <i>Num<sub>births</sub></i>    | 0.112***<br>(0.020)        | −0.047**<br>(0.018)     | 0.204***<br>(0.035)     | 0.011<br>(0.023)               | 0.0003<br>(0.020)            | 0.007<br>(0.012)              |
| <i>Num<sub>deaths</sub></i>    | 0.017<br>(0.027)           | 0.016<br>(0.031)        | 0.099**<br>(0.047)      | 0.041<br>(0.032)               | 0.018<br>(0.034)             | 0.067***<br>(0.023)           |
| <i>HH<sub>moved</sub></i>      | 0.538<br>(0.617)           | 1.900***<br>(0.695)     | 2.664***<br>(1.027)     | 0.427<br>(0.633)               | −0.972*<br>(0.571)           | 0.016<br>(0.337)              |
| Observations                   | 13,381                     | 13,381                  | 13,381                  | 13,381                         | 13,381                       | 13,381                        |
| R <sup>2</sup>                 | 0.638                      | −0.061                  | 0.555                   | 0.586                          | 0.299                        | 0.361                         |
| Adjusted R <sup>2</sup>        | 0.422                      | −0.695                  | 0.289                   | 0.339                          | −0.120                       | −0.020                        |
| <i>F<sup>Effective</sup></i>   | 19.5784                    | 19.5784                 | 19.5784                 | 19.5784                        | 19.5784                      | 19.5784                       |

Table 24: \*

Note: Outcome variables (1) - (3) represent the number of Pentecostal in the household. Outcomes (4) - (6) are the number of converts in the household. The instrument is exposure to concrete shocks. All regressions include wave and household fixed effects, and district\*wave fixed effects. Errors are clustered at the community level. Regression includes controls for income, education, mortality, fertility and marriage. \*p<0.1; \*\*p<0.05; \*\*\*p<0.01

## B.2 Out Migration

- The outmigration of household members leads to lower levels of Pentecostalism in the household.
- Outmigration reduces cross-denominational conversion.
  - Household leans more heavily on existing network.
  - If migration is church assisted, it may induce loyalty to the church.

Table 25:

|                                | <i>Dependent variable:</i> |                         |                         |                                |                              |                               |
|--------------------------------|----------------------------|-------------------------|-------------------------|--------------------------------|------------------------------|-------------------------------|
|                                | <i>Pent<sub>1</sub></i>    | <i>Pent<sub>2</sub></i> | <i>Pent<sub>3</sub></i> | <i>Convert<sub>denom</sub></i> | <i>Convert<sub>rel</sub></i> | <i>Convert<sub>Pent</sub></i> |
|                                | (1)                        | (2)                     | (3)                     | (4)                            | (5)                          | (6)                           |
| HH log(income)                 | 0.002<br>(0.002)           | 0.002<br>(0.002)        | 0.004<br>(0.003)        | 0.004*<br>(0.002)              | 0.001<br>(0.002)             | 0.001<br>(0.001)              |
| <i>Num<sub>Edu&gt;0</sub></i>  | 0.043*<br>(0.026)          | 0.166***<br>(0.027)     | 0.207***<br>(0.035)     | 0.182***<br>(0.040)            | 0.162***<br>(0.036)          | 0.049**<br>(0.019)            |
| <i>Num<sub>Primary</sub></i>   | 0.012<br>(0.023)           | -0.079***<br>(0.022)    | -0.040<br>(0.029)       | 0.028<br>(0.033)               | -0.009<br>(0.029)            | -0.029*<br>(0.017)            |
| <i>Num<sub>Secondary</sub></i> | -0.067**<br>(0.029)        | -0.016<br>(0.025)       | -0.069*<br>(0.038)      | -0.086**<br>(0.038)            | -0.037<br>(0.033)            | 0.020<br>(0.015)              |
| <i>Num<sub>Tertiary</sub></i>  | 0.110*<br>(0.064)          | -0.109*<br>(0.065)      | -0.035<br>(0.071)       | 0.084<br>(0.084)               | 0.056<br>(0.071)             | -0.036<br>(0.041)             |
| <i>Num<sub>Prof.Deg.</sub></i> | 0.036<br>(0.041)           | -0.062<br>(0.044)       | 0.005<br>(0.054)        | 0.092*<br>(0.049)              | -0.080*<br>(0.042)           | -0.034<br>(0.025)             |
| <i>Num<sub>Married</sub></i>   | 0.090***<br>(0.016)        | 0.062***<br>(0.012)     | 0.240***<br>(0.023)     | 0.156***<br>(0.021)            | 0.045**<br>(0.019)           | 0.006<br>(0.010)              |
| <i>Num<sub>births</sub></i>    | 0.110***<br>(0.018)        | -0.025**<br>(0.012)     | 0.208***<br>(0.027)     | -0.009<br>(0.020)              | -0.013<br>(0.018)            | 0.004<br>(0.011)              |
| <i>Num<sub>deaths</sub></i>    | 0.031<br>(0.027)           | 0.023<br>(0.024)        | 0.147***<br>(0.040)     | 0.079**<br>(0.035)             | 0.017<br>(0.035)             | 0.071***<br>(0.025)           |
| <i>Num<sub>mig</sub></i>       | -0.088*<br>(0.051)         | -0.104**<br>(0.049)     | -0.343***<br>(0.075)    | -0.211***<br>(0.060)           | 0.035<br>(0.059)             | -0.022<br>(0.035)             |
| Observations                   | 13,381                     | 13,381                  | 13,381                  | 13,381                         | 13,381                       | 13,381                        |
| R <sup>2</sup>                 | 0.644                      | 0.534                   | 0.666                   | 0.535                          | 0.481                        | 0.360                         |
| Adjusted R <sup>2</sup>        | 0.431                      | 0.256                   | 0.466                   | 0.258                          | 0.171                        | -0.023                        |
| <i>F<sup>Effective</sup></i>   | 92.058                     | 92.058                  | 92.058                  | 92.058                         | 92.058                       | 92.058                        |

Table 26: \*

Note: Outcome variables (1) - (3) represent the number of Pentecostal in the household. Outcomes (4) - (6) are the number of converts in the household. Instruments are exposure to concrete shocks and exposure to gold shocks. All regressions include wave and household fixed effects, and district\*wave fixed effects. Errors are clustered at the community level. Regression includes controls for income, education, mortality, fertility and marriage. \*p<0.1; \*\*p<0.05; \*\*\*p<0.01

## B.3 Individual Migration

- There is a weak effect of migration on Pentecostalism.
  - There is significant attrition of migrants. I control for this using propensity weighting.
  - It's difficult to draw any concrete conclusions from this, given the small sample size. (493 migrants across the 3 waves, 195 of these Pentecostal, 153 of these are converts. In total 4388 converts out of 20781)

Table 27:

|                            | <i>Dependent variable:</i> |                         |                              |                                |                               |
|----------------------------|----------------------------|-------------------------|------------------------------|--------------------------------|-------------------------------|
|                            | <i>Pent<sub>1</sub></i>    | <i>Pent<sub>2</sub></i> | <i>Convert<sub>rel</sub></i> | <i>Convert<sub>denom</sub></i> | <i>Convert<sub>Pent</sub></i> |
|                            | (1)                        | (2)                     | (3)                          | (4)                            | (5)                           |
| <i>HHlog(income)</i>       | 0.004*<br>(0.002)          | 0.001<br>(0.002)        | -0.007**<br>(0.003)          | -0.003<br>(0.003)              | -0.002<br>(0.002)             |
| <i>Primary</i>             | -0.006<br>(0.011)          | 0.024<br>(0.017)        | -0.038<br>(0.024)            | -0.007<br>(0.024)              | -0.006<br>(0.017)             |
| <i>Secondary</i>           | -0.004<br>(0.017)          | -0.035*<br>(0.018)      | 0.016<br>(0.020)             | -0.004<br>(0.023)              | -0.015<br>(0.016)             |
| <i>Tertiary</i>            | -0.002<br>(0.028)          | -0.007<br>(0.032)       | 0.067<br>(0.064)             | 0.085<br>(0.056)               | -0.003<br>(0.034)             |
| <i>ProfDeg.</i>            | -0.011<br>(0.016)          | 0.028*<br>(0.016)       | 0.028<br>(0.032)             | 0.053<br>(0.033)               | -0.008<br>(0.012)             |
| <i>Age</i>                 | 0.005<br>(0.004)           | 0.002<br>(0.003)        | -0.023***<br>(0.006)         | -0.014**<br>(0.006)            | -0.011***<br>(0.003)          |
| <i>Age<sup>2</sup></i>     | -0.00004<br>(0.00003)      | -0.00001<br>(0.00003)   | 0.0002***<br>(0.0001)        | 0.0001**<br>(0.00004)          | 0.0001***<br>(0.00003)        |
| <i>Num<sub>old</sub></i>   | -0.016<br>(0.011)          | 0.004<br>(0.008)        | 0.022<br>(0.015)             | 0.008<br>(0.012)               | -0.003<br>(0.009)             |
| <i>Num<sub>child</sub></i> | -0.002<br>(0.003)          | 0.0005<br>(0.003)       | -0.009<br>(0.007)            | 0.001<br>(0.005)               | -0.002<br>(0.003)             |
| <i>Married</i>             | 0.007<br>(0.017)           | 0.011<br>(0.018)        | -0.024<br>(0.021)            | -0.026<br>(0.022)              | 0.012<br>(0.014)              |
| <i>Migrate</i>             | 0.097<br>(0.141)           | 0.255*<br>(0.151)       | -0.459<br>(0.287)            | 0.018<br>(0.197)               | 0.026<br>(0.144)              |
| Observations               | 18,889                     | 18,889                  | 18,884                       | 18,884                         | 18,884                        |
| R <sup>2</sup>             | 0.876                      | 0.817                   | 0.599                        | 0.699                          | 0.515                         |
| Adjusted R <sup>2</sup>    | 0.762                      | 0.648                   | 0.228                        | 0.422                          | 0.067                         |
| <i>F<sup>Eff</sup></i>     | 32.0575                    | 32.0575                 | 32.0575                      | 32.0575                        | 32.0575                       |

Table 28: \*

Note: Outcome variables (1) and (2) are binary variables if the individual identifies as Pentecostal, or Renewalist, respectively. Outcomes (4) - (6) are dummy variables for Denominational conversion, religion conversion, and Pentecostal conversion respectively. Instruments are exposure to concrete shocks and exposure to gold shocks.<sup>36</sup> All regressions include wave and individual fixed effects, and district\*wave fixed effects. Errors are clustered at the district level. Regression includes controls for income, education, number of old, number of children, marriage status, age. Observations are weighted to account for attrition. \*p<0.1; \*\*p<0.05; \*\*\*p<0.01

## C Appendix: Single Instrument

Table 29:

|                                 | <i>Dependent variable:</i> |                          |                          |                                 |                               |                                 |
|---------------------------------|----------------------------|--------------------------|--------------------------|---------------------------------|-------------------------------|---------------------------------|
|                                 | <i>Pent</i> <sub>1</sub>   | <i>Pent</i> <sub>2</sub> | <i>Pent</i> <sub>3</sub> | <i>Convert</i> <sub>denom</sub> | <i>Convert</i> <sub>rel</sub> | <i>Convert</i> <sub>total</sub> |
|                                 | (1)                        | (2)                      | (3)                      | (4)                             | (5)                           | (6)                             |
| HH log(income)                  | 0.002<br>(0.002)           | 0.002<br>(0.002)         | 0.004<br>(0.003)         | 0.004*<br>(0.002)               | 0.002<br>(0.003)              | 0.004<br>(0.003)                |
| <i>Num</i> <sub>Edu&gt;0</sub>  | 0.039<br>(0.027)           | 0.180***<br>(0.027)      | 0.200***<br>(0.037)      | 0.159***<br>(0.041)             | 0.200***<br>(0.043)           | 0.041<br>(0.037)                |
| <i>Num</i> <sub>Primary</sub>   | 0.012<br>(0.023)           | -0.080***<br>(0.022)     | -0.039<br>(0.029)        | 0.029<br>(0.034)                | 0.004<br>(0.032)              | -0.039<br>(0.032)               |
| <i>Num</i> <sub>Secondary</sub> | -0.067**<br>(0.029)        | -0.019<br>(0.025)        | -0.068*<br>(0.038)       | -0.082**<br>(0.038)             | 0.008<br>(0.043)              | 0.008<br>(0.043)                |
| <i>Num</i> <sub>Tertiary</sub>  | 0.107*<br>(0.064)          | -0.099<br>(0.065)        | -0.040<br>(0.072)        | 0.068<br>(0.086)                | 0.043<br>(0.088)              | -0.040<br>(0.086)               |
| <i>Num</i> <sub>Prof.Deg.</sub> | 0.036<br>(0.041)           | -0.063<br>(0.044)        | 0.005<br>(0.055)         | 0.093*<br>(0.050)               | -0.089<br>(0.055)             | -0.089<br>(0.055)               |
| <i>Num</i> <sub>Married</sub>   | 0.094***<br>(0.016)        | 0.048***<br>(0.011)      | 0.247***<br>(0.024)      | 0.178***<br>(0.022)             | 0.020<br>(0.025)              | 0.020<br>(0.025)                |
| <i>Num</i> <sub>births</sub>    | 0.107***<br>(0.019)        | -0.013<br>(0.011)        | 0.202***<br>(0.028)      | -0.028<br>(0.022)               | -0.025<br>(0.023)             | 0.008<br>(0.023)                |
| <i>Num</i> <sub>deaths</sub>    | 0.036<br>(0.027)           | 0.004<br>(0.023)         | 0.157***<br>(0.041)      | 0.108***<br>(0.036)             | -0.002<br>(0.043)             | 0.075<br>(0.043)                |
| <i>Num</i> <sub>mig</sub>       | -0.115**<br>(0.054)        | -0.010<br>(0.047)        | -0.390***<br>(0.088)     | -0.360***<br>(0.069)            | 0.070<br>(0.090)              | -0.010<br>(0.090)               |
| <i>F</i> <sup>Eff</sup>         | 203.5715                   | 203.5715                 | 203.5715                 | 203.5715                        | 203.5715                      | 203.5715                        |
| Observations                    | 13,381                     | 13,381                   | 13,381                   | 13,381                          | 13,381                        | 13,381                          |
| R <sup>2</sup>                  | 0.644                      | 0.534                    | 0.666                    | 0.535                           | 0.481                         | 0.380                           |
| Adjusted R <sup>2</sup>         | 0.431                      | 0.256                    | 0.466                    | 0.258                           | 0.171                         | -0.011                          |
| Residual Std. Error (df = 8377) | 0.759                      | 0.558                    | 1.038                    | 0.895                           | 0.985                         | 0.430                           |

Table 30: \*

Note: Outcome variables (1) - (3) represent the number of Pentecostal in the household. Outcomes (4) - (6) are the number of converts in the household. Instruments are exposure to concrete shocks and exposure to gold shocks. All regressions include wave and household fixed effects, and district\*wave fixed effects. Errors are clustered at the community level. Regression includes controls for income, education, mortality, fertility and marriage. \*p<0.1; \*\*p<0.05; \*\*\*p<0.01

D Appendix: Household size

E Appendix: No Interacted Control

Table 31:

|                                | <i>Dependent variable:</i> |                         |                         |                                |                              |                               |
|--------------------------------|----------------------------|-------------------------|-------------------------|--------------------------------|------------------------------|-------------------------------|
|                                | <i>Pent<sub>1</sub></i>    | <i>Pent<sub>2</sub></i> | <i>Pent<sub>3</sub></i> | <i>Convert<sub>denom</sub></i> | <i>Convert<sub>rel</sub></i> | <i>Convert<sub>Pent</sub></i> |
|                                | (1)                        | (2)                     | (3)                     | (4)                            | (5)                          | (6)                           |
| HH log(income)                 | −0.00001<br>(0.002)        | 0.001<br>(0.002)        | 0.0005<br>(0.003)       | 0.001<br>(0.002)               | −0.002<br>(0.003)            | 0.001<br>(0.001)              |
| <i>Num<sub>Edu&gt;0</sub></i>  | 0.004<br>(0.026)           | 0.152***<br>(0.026)     | 0.137***<br>(0.033)     | 0.124***<br>(0.036)            | 0.115***<br>(0.038)          | 0.039**<br>(0.019)            |
| <i>Num<sub>Primary</sub></i>   | 0.015<br>(0.023)           | −0.078***<br>(0.022)    | −0.035<br>(0.029)       | 0.031<br>(0.032)               | 0.010<br>(0.031)             | −0.028*<br>(0.017)            |
| <i>Num<sub>Secondary</sub></i> | −0.056**<br>(0.028)        | −0.012<br>(0.025)       | −0.049<br>(0.038)       | −0.070*<br>(0.037)             | 0.033<br>(0.042)             | 0.023<br>(0.016)              |
| <i>Num<sub>Tertiary</sub></i>  | 0.113*<br>(0.065)          | −0.108*<br>(0.065)      | −0.029<br>(0.073)       | 0.088<br>(0.085)               | 0.058<br>(0.090)             | −0.035<br>(0.041)             |
| <i>Num<sub>Prof.Deg.</sub></i> | 0.046<br>(0.041)           | −0.058<br>(0.044)       | 0.023<br>(0.055)        | 0.107**<br>(0.049)             | −0.065<br>(0.055)            | −0.031<br>(0.025)             |
| <i>Num<sub>Married</sub></i>   | 0.076***<br>(0.016)        | 0.057***<br>(0.013)     | 0.214***<br>(0.024)     | 0.135***<br>(0.021)            | −0.025<br>(0.025)            | 0.003<br>(0.010)              |
| <i>Num<sub>births</sub></i>    | 0.043**<br>(0.018)         | −0.049***<br>(0.012)    | 0.087***<br>(0.026)     | −0.109***<br>(0.023)           | −0.181***<br>(0.028)         | −0.014<br>(0.011)             |
| <i>Num<sub>deaths</sub></i>    | 0.041<br>(0.027)           | 0.026<br>(0.024)        | 0.165***<br>(0.040)     | 0.094***<br>(0.035)            | 0.009<br>(0.042)             | 0.074***<br>(0.024)           |
| <i>HHldSize</i>                | 0.052***<br>(0.009)        | 0.018***<br>(0.006)     | 0.092***<br>(0.013)     | 0.076***<br>(0.012)            | 0.126***<br>(0.015)          | 0.014**<br>(0.006)            |
| <i>Num<sub>mig</sub></i>       | −0.117**<br>(0.048)        | −0.112**<br>(0.046)     | −0.394***<br>(0.069)    | −0.255***<br>(0.057)           | 0.067<br>(0.080)             | −0.030<br>(0.034)             |
| Observations                   | 13,381                     | 13,381                  | 13,381                  | 13,381                         | 13,381                       | 13,381                        |
| R <sup>2</sup>                 | 0.646                      | 0.529                   | 0.670                   | 0.569                          | 0.497                        | 0.362                         |
| Adjusted R <sup>2</sup>        | 0.435                      | 0.247                   | 0.473                   | 0.312                          | 0.197                        | −0.019                        |
| <i>F<sup>Effective</sup></i>   | 92.058                     | 92.058                  | 92.058                  | 92.058                         | 92.058                       | 92.058                        |

Table 32: \*

Note: Outcome variables (1) - (3) represent the number of Pentecostal in the household. Outcomes (4) - (6) are the number of converts in the household. Instruments are exposure to concrete shocks and exposure to gold shocks. All regressions include wave and household fixed effects, and district\*wave fixed effects. Errors are clustered at the community level. Regression

Table 33:

|                              | <i>Dependent variable:</i> |                          |                          |                                |                               |
|------------------------------|----------------------------|--------------------------|--------------------------|--------------------------------|-------------------------------|
|                              | <i>Pent</i> <sub>1</sub>   | <i>Pent</i> <sub>2</sub> | <i>Pent</i> <sub>3</sub> | <i>Convert</i> <sub>pent</sub> | <i>Convert</i> <sub>any</sub> |
|                              | (1)                        | (2)                      | (3)                      | (4)                            | (5)                           |
| <i>exp<sub>gold</sub></i>    | 1.013<br>(3.119)           | −1.373<br>(2.971)        | −0.666<br>(2.379)        | 4.111<br>(2.716)               | 5.799<br>(5.292)              |
| <i>exp<sub>cement</sub></i>  | −49.546<br>(51.532)        | 30.101<br>(65.229)       | −39.711<br>(61.255)      | −96.007***<br>(35.467)         | 131.604<br>(80.253)           |
| <i>HH<sub>move</sub></i>     | −2.688**<br>(1.191)        | −2.835**<br>(1.100)      | −1.891<br>(1.194)        | −2.181***<br>(0.788)           | 2.172<br>(1.557)              |
| Observations                 | 16,931                     | 16,931                   | 16,931                   | 16,853                         | 16,853                        |
| R <sup>2</sup>               | 0.545                      | 0.389                    | 0.753                    | −0.508                         | 0.542                         |
| Adjusted R <sup>2</sup>      | 0.370                      | 0.153                    | 0.658                    | −1.091                         | 0.365                         |
| <i>F<sup>Effective</sup></i> | 10.4113                    | 10.4113                  | 10.4113                  | 10.4113                        | 10.4113                       |

Table 34: \*

Note: All regressions include time and household fixed effects. Errors are clustered at the community level. \*p<0.1; \*\*p<0.05; \*\*\*p<0.01

Table 35:

|                         | <i>Dependent variable:</i> |                        |                    |                     |                    |
|-------------------------|----------------------------|------------------------|--------------------|---------------------|--------------------|
|                         | $Pent_1$                   | $Pent_2$               | $Pent_3$           | $Convert_{pent}$    | $Convert_{any}$    |
|                         | (1)                        | (2)                    | (3)                | (4)                 | (5)                |
| $exp_{gold}$            | -1.409<br>(1.282)          | -2.915<br>(1.850)      | -2.217<br>(1.750)  | 0.964<br>(1.297)    | 7.104*<br>(4.211)  |
| $exp_{cement}$          | 38.849<br>(33.123)         | 123.763***<br>(43.624) | 22.491<br>(43.394) | -23.432<br>(16.793) | 58.872<br>(50.418) |
| $HH_{mig}$              | 0.214***<br>(0.076)        | 0.192**<br>(0.083)     | 0.171*<br>(0.088)  | 0.181***<br>(0.056) | -0.230*<br>(0.127) |
| Observations            | 16,927                     | 16,927                 | 16,927             | 16,853              | 16,853             |
| R <sup>2</sup>          | 0.750                      | 0.638                  | 0.799              | 0.293               | 0.658              |
| Adjusted R <sup>2</sup> | 0.654                      | 0.499                  | 0.721              | 0.019               | 0.525              |
| $F^{Effective}$         | 51.7698                    | 51.7698                | 51.7698            | 51.7698             | 51.7698            |

Table 36: \*

Note: All regressions include time and household fixed effects. Errors are clustered at the community level. \*p<0.1; \*\*p<0.05; \*\*\*p<0.01

## F Appendix: Using exposure as a control

Table 37:

|                                    | <i>Dependent variable:</i> |                          |                          |                                |                               |
|------------------------------------|----------------------------|--------------------------|--------------------------|--------------------------------|-------------------------------|
|                                    | <i>Pent</i> <sub>1</sub>   | <i>Pent</i> <sub>2</sub> | <i>Pent</i> <sub>3</sub> | <i>Convert</i> <sub>pent</sub> | <i>Convert</i> <sub>any</sub> |
|                                    | (1)                        | (2)                      | (3)                      | (4)                            | (5)                           |
| <i>exp<sub>gold</sub></i>          | −0.004<br>(2.496)          | 1.478<br>(2.783)         | 0.010<br>(2.265)         | 1.139<br>(2.368)               | 6.255<br>(4.462)              |
| <i>outsideexp<sub>gold</sub></i>   | −4.170<br>(22.469)         | −63.548**<br>(27.911)    | −12.676<br>(26.044)      | 26.248<br>(24.155)             | −20.822<br>(40.909)           |
| <i>exp<sub>cement</sub></i>        | −107.439*<br>(56.626)      | 14.596<br>(79.934)       | 11.961<br>(76.527)       | −82.285**<br>(40.435)          | 97.353<br>(74.719)            |
| <i>outsideexp<sub>cement</sub></i> | 145.085**<br>(69.687)      | 36.138<br>(103.584)      | −123.110<br>(102.415)    | −43.095<br>(55.788)            | 57.364<br>(67.816)            |
| <i>HH<sub>move</sub></i>           | −2.368**<br>(1.194)        | −3.042**<br>(1.222)      | −2.148<br>(1.310)        | −2.268**<br>(0.890)            | 1.897<br>(1.576)              |
| Observations                       | 16,931                     | 16,931                   | 16,931                   | 16,853                         | 16,853                        |
| R <sup>2</sup>                     | 0.599                      | 0.348                    | 0.736                    | −0.579                         | 0.572                         |
| Adjusted R <sup>2</sup>            | 0.444                      | 0.097                    | 0.635                    | −1.190                         | 0.407                         |
| <i>F<sup>Effective</sup></i>       | 9.5124                     | 9.5124                   | 9.5124                   | 9.5124                         | 9.5124                        |

Table 38: \*

Note: All regressions include wave and household fixed effects. Regression includes a control for exposure to surrounding districts (*outsideexp<sub>gold</sub>*, *outsideexp<sub>cement</sub>*). Errors are clustered at the community level. \*p<0.1; \*\*p<0.05; \*\*\*p<0.01

Table 39:

|                                    | <i>Dependent variable:</i> |                         |                         |                               |                              |
|------------------------------------|----------------------------|-------------------------|-------------------------|-------------------------------|------------------------------|
|                                    | <i>Pent<sub>1</sub></i>    | <i>Pent<sub>2</sub></i> | <i>Pent<sub>3</sub></i> | <i>Convert<sub>pent</sub></i> | <i>Convert<sub>any</sub></i> |
|                                    | (1)                        | (2)                     | (3)                     | (4)                           | (5)                          |
| <i>exp<sub>gold</sub></i>          | −2.263*<br>(1.220)         | −1.237<br>(2.023)       | −2.157<br>(1.848)       | −1.013<br>(1.342)             | 8.459*<br>(4.540)            |
| <i>outsideexp<sub>gold</sub></i>   | 13.769<br>(15.936)         | −38.838**<br>(19.236)   | 2.382<br>(20.522)       | 44.465**<br>(21.105)          | −31.924<br>(42.018)          |
| <i>exp<sub>cement</sub></i>        | −37.276<br>(41.225)        | 103.293<br>(67.596)     | 76.640<br>(63.993)      | −13.023<br>(29.692)           | 36.202<br>(49.036)           |
| <i>outsideexp<sub>cement</sub></i> | 156.517**<br>(64.143)      | 53.928<br>(92.465)      | −114.728<br>(96.348)    | −34.066<br>(46.577)           | 56.396<br>(65.807)           |
| <i>HH<sub>mig</sub></i>            | 0.183**<br>(0.074)         | 0.197**<br>(0.086)      | 0.190**<br>(0.089)      | 0.172***<br>(0.057)           | −0.227*<br>(0.126)           |
| Observations                       | 16,927                     | 16,927                  | 16,927                  | 16,853                        | 16,853                       |
| R <sup>2</sup>                     | 0.757                      | 0.638                   | 0.797                   | 0.301                         | 0.658                        |
| Adjusted R <sup>2</sup>            | 0.664                      | 0.498                   | 0.719                   | 0.031                         | 0.526                        |
| <i>F<sup>Effective</sup></i>       | 48.5155                    | 48.5155                 | 48.5155                 | 48.5155                       | 48.5155                      |

Table 40: \*

Note: All regressions include wave and household fixed effects. Regression includes a control for exposure to surrounding districts (*outsideexp<sub>gold</sub>*, *outsideexp<sub>cement</sub>*). Errors are clustered at the community level. \*p<0.1; \*\*p<0.05; \*\*\*p<0.01

## G Appendix: Using self reported migration

Table 41:

|                               | <i>Dependent variable:</i> |                          |                          |                                |
|-------------------------------|----------------------------|--------------------------|--------------------------|--------------------------------|
|                               | <i>Pent</i> <sub>1</sub>   | <i>Pent</i> <sub>2</sub> | <i>Pent</i> <sub>3</sub> | <i>Convert</i> <sub>pent</sub> |
|                               | (1)                        | (2)                      | (3)                      | (4)                            |
| <i>exp</i> <sub>steel</sub>   | −7.501<br>(8.698)          | −5.769<br>(5.548)        | −7.510<br>(7.586)        | −2.786<br>(6.052)              |
| ‘hhmig(fit)’                  | −4.934<br>(4.328)          | −2.988<br>(2.832)        | −4.202<br>(3.835)        | −3.325<br>(3.058)              |
| <i>F</i> <sup>Effective</sup> | 1.6945                     | 1.6945                   | 1.6945                   | 1.6945                         |
| Observations                  | 15,425                     | 15,425                   | 15,425                   | 15,384                         |
| R <sup>2</sup>                | −0.593                     | 0.145                    | 0.312                    | −3.119                         |
| Adjusted R <sup>2</sup>       | −1.127                     | −0.141                   | 0.082                    | −4.503                         |
| Residual Std. Error           | 1.766 (df = 11557)         | 1.275 (df = 11557)       | 1.641 (df = 11557)       | 1.203 (df = 1151)              |

Note:

\*p<

Table 42:

|                               | <i>Dependent variable:</i> |                          |                          |                                |
|-------------------------------|----------------------------|--------------------------|--------------------------|--------------------------------|
|                               | <i>Pent</i> <sub>1</sub>   | <i>Pent</i> <sub>2</sub> | <i>Pent</i> <sub>3</sub> | <i>Convert</i> <sub>pent</sub> |
|                               | (1)                        | (2)                      | (3)                      | (4)                            |
| <i>exp</i> <sub>gold</sub>    | −1.073<br>(4.029)          | −1.876<br>(2.601)        | −2.035<br>(3.484)        | 1.560<br>(2.980)               |
| <i>HH</i> <sub>immig</sub>    | 2.407<br>(1.745)           | 1.457<br>(1.210)         | 2.050<br>(1.631)         | 1.629<br>(1.257)               |
| <i>F</i> <sup>Effective</sup> | 2.2802                     | 2.2802                   | 2.2802                   | 2.2802                         |
| Observations                  | 15,425                     | 15,425                   | 15,425                   | 15,384                         |
| R <sup>2</sup>                | 0.329                      | 0.492                    | 0.642                    | −0.857                         |
| Adjusted R <sup>2</sup>       | 0.105                      | 0.322                    | 0.522                    | −1.480                         |
| Residual Std. Error           | 1.146 (df = 11557)         | 0.983 (df = 11557)       | 1.184 (df = 11557)       | 0.807 (df = 1151)              |

Note:

\*p<